NAVEEN RAJ M

Mechanical Design Engineer | MV/LV Switchgear Design (IEC 62271 / IEC 61439) | 11–33 kV AIS Panels | Sheet Metal Engineering | SolidWorks | Solid Edge | Inventor

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PROFESSIONAL SUMMARY

Mechanical Design Engineer with 3+ years of experience in MV (IEC 62271) and LV (IEC 61439) electrical panel engineering, from concept development to production-ready drawings. R&D experience on 11 kV, 22 kV and 33 kV Air Insulated Switchgear (AIS) panels for indoor and outdoor installation, including rack-in/rack-out mechanisms, automatic shutter assemblies, and clearance-compliant placement of control transformers (CTs) and potential transformers (PTs). Design experience across TI, ENERSYS, Siemens 8PU SIEPAN, SIEPAN ELITE and Siemens IPAN panels, with focus on short-circuit withstand capability, IP protection (IEC 60529), internal arc-fault containment (IEC 62271-200 for MV, IEC TR 61641 for LV) and seismic withstand (IS 1893), External Mechanical Impact (IEC 62262). Strong expertise in sheet metal design, GD&T, weldment design, tolerance stack-up analysis, forming allowances and DFM/DFA practices. Experienced in busbar mechanical strength assessment, thermal performance and heat dissipation analysis as per IEC 61439-1 and IEC TR 60890 (airflow optimization, CFM calculation, VFD panel ventilation), enclosure structural validation, and creepage and clearance optimization. Well-versed in Form of Separation (Form 1 to Form 4). Proficient in SolidWorks, Solid Edge and AutoCAD (2D/3D), with parametric 3D modeling, FEA-supported design refinement, detailed 2D drafting and BOM generation.

KEY ACHIEVEMENTS

- Contributed to R&D of 11/22/33 kV AIS panels (indoor and outdoor), covering panel construction, rack-in/rack-out mechanisms and automatic shutter assemblies.
- Designed 500+ LV panel variants, reducing design revision cycles and improving manufacturing efficiency.
- Delivered sheet metal designs for MV/LV panels including VCB, APFC, IMCC, MCC, PCC, MPDB, MLDB, MSB, LSB, MLSB, SSB, AMF, and PDB across multiple projects.
- Improved fabrication accuracy and reduced assembly rework through GD&T and tolerance stack-up analysis.
- Contributed to R&D activities including prototyping and testing, enhancing enclosure thermal efficiency and IP compliance.

CORE COMPETENCIES & TECHNICAL SKILLS

Design Standards	IEC 61439 Low Voltage (LV) • IEC 62271 Medium Voltage (MV) • IP Protection (IEC 60529) • Internal Arc-Fault Containment (IEC 62271-200 / IEC TR 61641) • Seismic Withstand (IS 1893) • Short-Circuit Withstand Capability • Creepage / Clearance Layout • Busbar Sizing & Mechanical Assessment • Thermal & Heat Load Calculations (IEC 61439-1 / IEC TR 60890) • Enclosure Structural Validation & Load Calculations • Modular & Semi-Modular Panel Design
MV Panel Design	11/22/33 kV AIS (Indoor & Outdoor) • Rack-in/Rack-out Mechanisms • Automatic Shutters • CT/PT Mounting & Clearances
Product Platforms	TI • ENERSYS • Siemens 8PU SIEPAN • SIEPAN ELITE • Siemens IPAN
Software / CAD	SolidWorks (Advanced) • Solid Edge (Advanced) • Inventor (Advanced) • AutoCAD 2D & 3D (Advanced) • ANSYS FEA (Structural & Thermal) • 3D Printing & Rapid Prototyping
Manufacturing	Sheet Metal Design & DFM/DFA • Forming Allowances • GD&T & Tolerance Stack-Up Analysis • Weldment Design • Punching & Laser CNC Programming • Jig & Fixture Design • Engineering Drawing & Detailing • BOM Documentation & Release

PROFESSIONAL EXPERIENCE

Hertz Panels (P) Ltd – Hosur, Tamil Nadu — April 2025 – Present

Mechanical Design Engineer

- Design and develop IEC-compliant LV and MV electrical panel enclosures for TI, ENERSYS, Siemens 8PU SIEPAN, SIEPAN ELITE and Siemens IPAN product lines using SolidWorks and AutoCAD.
- R&D and design of 11 kV, 22 kV and 33 kV AIS panels for indoor and outdoor installation, including rack-in/rack-out mechanisms, automatic shutter assemblies, and clearance-compliant placement of control transformers and potential transformers.
- Perform sheet metal design for various switchgear panel types, including 3D modelling, detailed engineering drawings, and manufacturing BOMs.
- Conduct thermal performance validation (temperature rise as per IEC 61439-1 and IEC TR 60890), IP protection verification (IEC 60529), seismic withstand checks (IS 1893), External Mechanical Impact (IEC 62262) and IEC 61439/62271 design checks to confirm LV and MV enclosures meet specified withstand and protection requirements.
- Evaluate busbar mechanical assessment, creepage/clearance layout, and internal arc-fault containment (IEC 62271-200 for MV, IEC TR 61641 for LV) to ensure electrical safety and durability per IEC 62271 requirements.
- Coordinate with electrical and production teams using components from ABB, L&K, Siemens, and Schneider Electric to ensure design-to-manufacturing feasibility.

BK Power HT & LT Switch Board Mfg. – Coimbatore, Tamil Nadu Nov 2023 – April 2025 (1.6 Years)

Mechanical Design Engineer

- Designed MV and LV panel enclosures in sheet metal for a wide range of panel types: HT VCB, HT Capacitor, APFC, MV, AMF, MSB, LSB, MLSB, SSB, PCC, MCC, and PDB.
- Ensured all MV panel designs conformed to IEC 62271 (Medium-Voltage Switchgear) and LV assemblies met IEC 61439 compliance requirements for internal arc (IEC 62271-200 for MV, IEC TR 61641 for LV), IP rating (IEC 60529), seismic withstand (IS 1893), and short-circuit withstand.
- Created detailed 3D models and 2D manufacturing drawings using Solid Edge and AutoCAD, ensuring compliance with customer specifications and IEC standards.
- Applied GD&T standards and tolerance stack-up analysis to ensure dimensional accuracy and reduce fabrication errors across assemblies.
- Managed complete BOM documentation including part lists, hardware specifications, and revision control for multiple concurrent panel projects.
- Utilized Punching and Laser software for sheet metal programming, nesting optimization, and CNC punching to improve material utilization.
- Worked with components from ABB, L&K, Legrand, and Siemens; coordinated with procurement and fabrication teams on material selection and design-for-manufacturing improvements.
- Supported R&D activities including enclosure prototyping and performance testing to validate design changes before production release.

INTERNSHIP EXPERIENCE

Schneider Electric India (P) Ltd – Chennai, Tamil Nadu April 2023 (1 Month)

- Observed and supported manufacturing operations for MCB and RCCB product lines at a global-standard facility, gaining insight into quality control, production planning, and IEC-compliant manufacturing processes.

Venus Home Appliances (P) Ltd – Thoothukudi, Tamil Nadu November 2022 (2 Weeks)

- Gained hands-on exposure to electro-mechanical assembly, quality inspection, and production workflows for water heaters and ceiling fans.

EDUCATION

B.E. – Mechanical Engineering | CGPA: 8.22 / 10 2020 – 2023

St. Mother Theresa Engineering College, Thoothukudi

- Mini Project: Design and Fabrication of Portable Gas Cylinder Trolley – designed structural components and fabrication drawings
- Research Project: Tribological Properties of Mg-ZrC Composites – investigated wear behaviour of Magnesium (Mg)-Zirconium carbide (ZrC) composite materials

Diploma – Mechanical Engineering | Average: 69% 2016 – 2019

Jeya Polytechnic College, Thoothukudi

- Diploma Project: Design and Fabrication of Pedal Power Water Pump – designed mechanical drive system and fabricated working prototype

RESEARCH PUBLICATION

Paper Title: "Effects of Polarity Parameter on Machining of Tool Steel Workpiece Using Electrical Discharge Machining (EDM)"

Investigated the effect of polarity on EDM material removal rate (MRR) and surface finish, contributing to optimized machining parameters for tool steel.

PROFESSIONAL PROFILE

Languages: Tamil (Native), English (Professional)	Availability: Immediate
Work Authorization: India	Willing to Relocate: Yes